IDS 705 Final Report

Project Title

Subtitle [as applicable]

Team Members:
Name 1
Name 2
Name 3
Name 4
[in alphabetical order by last name]

Team Identifier: ####

Target stakeholder(s) for this report: []

Please follow the Data Science Memo instructions here, and much of the content below comes directly from that document. Any instructions in THIS document supersede the instructions in the link above and are specific to this course. This document should be between 4 and 6 pages (excluding the appendices).

Executive Summary

[1-2 paragraphs] This should be the one paragraph that captures the significance of what you did and why you did it - this should be a summary of the work ***and your outcomes and conclusions*** in brief. The Executive Summary should contain everything a reader needs to know if they were to read only the summary. It should fully summarize the problem that motivates the project, what the project intends to do, how what it intends to do is meant to help address the problem, and the key takeaway of the analysis.

Please see this discussion for more on the content of this section.

Decisions to be made: After the summary, most stakeholders appreciate being told the critical decisions about next steps that need to be made based on the memo content. This can also be paired with a concise summary of the information you believe is critical for them to know when making their decisions. This helps stakeholders understand what they should focus on as they read, and (as discussed below) helps ensure you get across your most important points to your stakeholder even if they don't finish your memo.

See this discussion for more on the content of this section. This can be a series of bullet points, for example.

Report

After the executive summary comes the body of the report. This section should focus on presenting each of the outcomes of your work that are important to the stakeholder you identified, starting with the most important. Each of the assertions you make should be supported with evidence, which is likely to be primarily in the form of figures and tables that support your most salient points. While brief mentions of what you did to contextualize the results will likely be necessary, detailed discussion of your methods is not appropriate here and should be saved for the appendix.

Please read more here for guidance on how to approach this section (and importantly, what not to do).

Required Appendices

Background and related work. One of the most important things that you can do when approaching a problem is to first read as much as you possibly can about how others have approached it in the past. It allows you to often leapfrog past bad initial ideas to the cutting edge of what is being done in the field. This is a superpower that will set you apart from those who choose to remain uninformed with inferior results. To that end, in this section, describe what you learned from what you read in this space and, importantly, how it informed your choices. This should not simply be a list of articles, but should synthesize what you learned from them into a coherent whole. At the end of this section, please include a list of references. A minimum of 15 references is required (at least 10 must be technical papers/reports or conference papers, rather than blogs or websites). Citations should follow a known model (MLA, IEEE, Chicago, APA, etc.). Citations SHOULD NOT JUST BE HYPERLINKS. When in doubt, use MLA.

Methods and experimental design. Present relevant details on your machine learning experiments (for supervised learning, a description of any preprocessing, feature extraction, classification/regression techniques, experimental designs, and evaluation criteria) and why you made each of the choices you did to achieve your goal. Also include a flow chart (or multiple flow charts if necessary) of your methodology so the reader can easily conceptualize your solution. The flow chart of the overall experimental design should clearly articulate your process (example).

Ethical Considerations. Relevant ethical considerations should be included directly in the memo itself. However, this section is meant to put a finer point on the topic.

Discuss any ethical aspects or implications of your project. What potential sources of bias have you encountered? What (if any) potential privacy concerns are associated with your project? Is there any potential for negative societal impact from the work (who benefits, who could be harmed, and what steps could you take to minimize negative consequences)? Are there any issues around transparency and explainability for your work (e.g., is that important for your application), and if so, how are you planning to address them?

Roles. Since this is a team project, we want to know what your specific contribution was to this project. Provide details on your individual role. Each team member should clearly articulate an individual role.

Optional Appendices

You may add as many additional components to the appendix as you like, and there are no length requirements there, although that content may not be read in evaluating the report, so make sure the body of the report contains the most important content.

Instructions: template and submission

*Note: this section is for information purposes and should **not** be included in your report*

How to use this template

1. Click File > Make a copy and delete the instructions, replacing text with your report content
2. Follow the formatting instructions contained in this document
3. Share your completed version of the file, giving general access to “anyone with the link” and at least “commenter” privileges, so that we can add comments and suggestions directly to the text

Headers

The main sections of the report should have the “Heading 1” style applied (as “Formatting” is shown here). Subsections, like this one, should have the “Heading 2” style applied and so on. This allows the report to be easily navigated.

Tables

Tables should have captions before the start of the table, and, like Table 1 below, should be referenced in the text.

Table 1. Experimental conditions investigated in this study. In this case, each experimental condition is listed in each row with the columns discussing the factors that differ across experiments

Experimental Conditions	**Factor 1 (e.g., hyperparameters, model types, training datasets, etc.)**	**Factor 2**	**Factor 3**
Condition 1			
Condition 2			
Condition 3			

Figures

Figures are highly encouraged, and should each be referenced in the text (such that every figure has a clear point to the story that you tell). Every figure should have a caption, figure number, axis labels (with units if applicable), and legend or direct labeling of data (as shown in Figure 1), if applicable. If you use any figures that are not your own, they should be cited as well. No figure should be superfluous - every figure should be referenced in the text. Figure 1, below, shows an example of a clear figure with a descriptive caption.

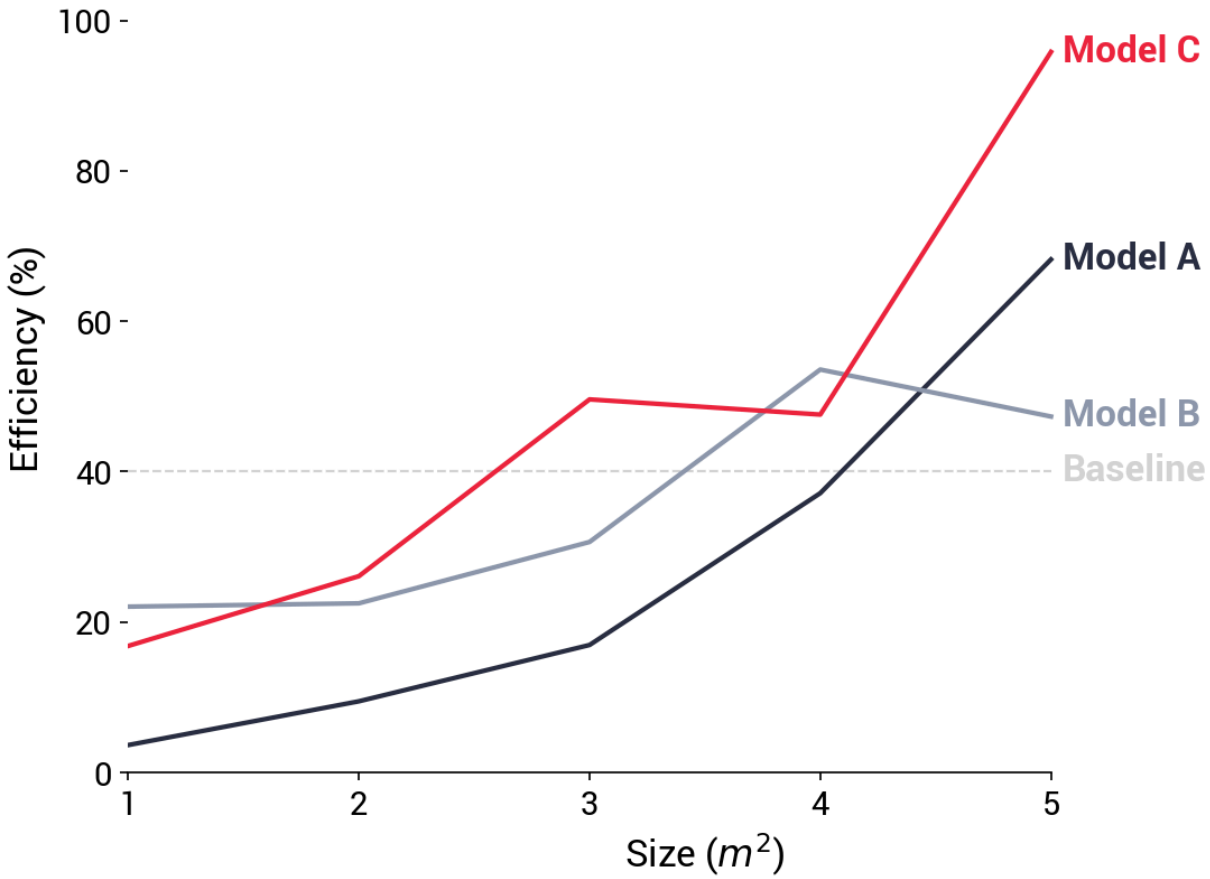

Figure 1. Experimental results show that Model C tends to outperform other models compared for the same training dataset.

Flowcharts

Every report should have a flowchart as described above. Some examples of this include:

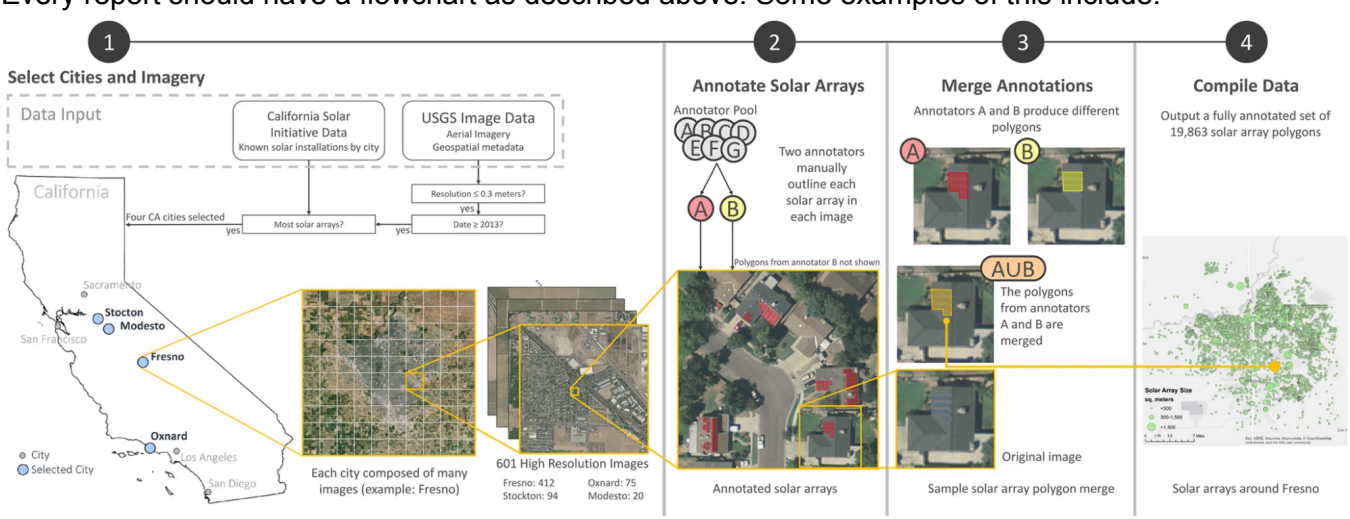

Figure 2. Example flowchart from Bradbury et al. 2016.

Rubric

Much of this comes directly from the MIDS Memo format, with some modifications for this course.

To aid with grading scalability, all rubric items in this common rubric may take on one of only three discrete values: 1, 0.5, or 0.

A. Does the memo clearly state the problem that motivates the analysis?

- 1: The memo clearly and explicitly states the problem (or all problems, if more than one) motivating the work presented in the memo. The problem is stated in concrete terms, and is a problem that the declared stakeholder would view as a problem over which they have potential agency.
- 0.5: Any of:
    - The work in the memo appears to be motivated by a specific problem, but the problem (problems) is (are) not clearly or explicitly stated.
    - The problem is explicitly stated, but at a level of generality that it is unclear the stakeholder would feel it is a problem they are in a position to solve ("Large levels of emissions are causing global warming." is an explicitly stated problem, but not one that any one stakeholder is likely to feel they have agency to solve, especially based on the work in one memo.)
- 0: The problem motivating the memo is unclear — not only is it not explicitly stated, but a clear motivating problem is also not obvious to the reader.

B. Does the memo clearly state what it intends to do and how that plan will help address the motivating problem?

- 1: The memo clearly and explicitly lays out how the plan of action/analysis for the memo will help address the stakeholder's problem. The logic is clear, and does not omit any large steps. Moreover, the steps seem reasonable given the stakeholder to whom the memo is addressed.
- 0.5:
    - One can imagine how the memo's contents could be used to address the problem, but the logic of how that might be done is not laid out clearly or explicitly
    - The memo lays out a way that it's contents could be used to help address the problem, but the logic of the reasoning strains credulity
- 0: How the memo content is meant to help solve the problem is unclear.

C. Does the memo start with a short *Executive Summary* section that states the motivating problem, what the analysis intends to accomplish, how that plan will help address the motivating problem, and the main takeaway of the analysis?

- 1: Yes, the memo includes an Executive Summary and it does all these tasks.
- 0.5: There is an Executive Summary, but while it summarizes the memo, it doesn't clearly and explicitly connect the dots from problem to strategy for solving problem to conclusions and take-aways.
- 0: There is no Executive Summary or it really doesn't meet its objectives.

D. For figures in the memo:

Are all figures clearly labelled and titled?

Is it clear what each figure in the memo is intended to convey? All figures should have a point they want to communicate to the reader — are those points clear?

Do all figures serve a clear purpose in the memo?

- 1: Yes, all figures achieve these goals.
- 0.5: These goals are sometimes met and sometimes not met, or met for some figures but not others.
- 0: These goals are being met less often than they are not being met.

E. Does the background appendix section clearly demonstrate a significant review of work related to the project topic and a connection to the choices of the project?

- 1: Yes, the background section demonstrates meaningful review of relevant work (primarily from professional reports, journal papers, conference publications, and the like) and a synthesis of what has been done as it relates to this project. This synthesis clearly impacted the project and was not an after-the-fact addition of papers to satisfy the requirement but demonstrates a commitment to understanding work that was done in this space.
- 0.5: Some works are listed that are relevant, but any of the following are true:
    - There are not many previous works referenced
    - There is little synthesis of those works to indicate comprehension
    - There are few, if any, tie-ins to the work or choices of the project
- 0: Few works are listed and show limited effort in review and influence in the direction of the project.

F. Do the methods and experimental design appendix clearly articulate the purpose of each experiment and the conditions of each experiment, supporting each design choice with clear, reasonable rationale? Are the choices made aligned with best practices and demonstrate a clear understanding of the application domain and the methods being used? Is there a flowchart that supports and articulates the work in the project?

- 1: Yes, this section achieves these goals.
- 0.5: These goals are sometimes met and sometimes not met.
- 0: These goals are being met less often than they are not being met.

G. Does the ethical considerations appendix meaningfully discuss the ethical considerations of the problem?

- 1: Yes, this section achieves this goal.
- 0.5: This section is attempted but misses key considerations around ethics for this problem or fails to adequately address how to respond to those considerations.
- 0: This section is skipped without justification for why ethical considerations may not be meaningful for this problem.